

Table S7 Regression: Efficiency and Buyer-Seller Disparity

|                   | $ P_{\tau} - P_{CE} $ |                      | Realized Surplus (%) |                      | Buyer-Seller Disparity     |                            |
|-------------------|-----------------------|----------------------|----------------------|----------------------|----------------------------|----------------------------|
|                   | T1-T20<br>(1)         | T11-T20<br>(2)       | T1-T20<br>(3)        | T11-T20<br>(4)       | T1-T20<br>(5)              | T11-T20<br>(6)             |
| Intercept         | 8.654***<br>(0.403)   | 8.976***<br>(0.610)  | 0.537***<br>(0.039)  | 0.666***<br>(0.048)  | 10506.774***<br>(1041.050) | 14420.906***<br>(1501.856) |
| Flow30            | -0.018<br>(0.073)     | -0.030<br>(0.093)    | -0.025***<br>(0.006) | -0.025***<br>(0.007) | -1607.811***<br>(161.163)  | -2082.046***<br>(207.071)  |
| Flow60            | 0.150**<br>(0.073)    | 0.246**<br>(0.096)   | 0.057***<br>(0.005)  | 0.047***<br>(0.006)  | -1613.308***<br>(148.599)  | -2297.966***<br>(188.476)  |
| Period number     | -0.255***<br>(0.021)  | -0.271***<br>(0.032) | 0.019***<br>(0.002)  | 0.012***<br>(0.002)  | -90.099<br>(56.335)        | -280.793***<br>(82.415)    |
| Interval number   | -0.019***<br>(0.002)  | -0.022***<br>(0.002) |                      |                      |                            |                            |
| Observations      | 18000                 | 9000                 | 3000                 | 1500                 | 3000                       | 1500                       |
| Adjusted $R^2$    | 0.359                 | 0.466                | 0.377                | 0.218                | 0.804                      | 0.833                      |
| CDA avg.          | 2.17                  | 2.13                 | 0.14                 | 0.13                 | 3.98                       | 3.88                       |
| Flow30 = Flow60 ? | 0.000                 | 0.000                | 0.000                | 0.000                | 0.973                      | 0.308                      |

### S1.4.3 Summary Statistics with Unweighted Prices

This table presents summary statistics using unweighted second-by-second price observations, rather than volume-weighted prices used in the main analysis.

Table S8 Summary Statistics of Experimental Sessions

|                             | T1 - T20 |        |        | T1 - T10 |        |        | T11 - T20 |        |        |
|-----------------------------|----------|--------|--------|----------|--------|--------|-----------|--------|--------|
|                             | CDA      | Flow30 | Flow60 | CDA      | Flow30 | Flow60 | CDA       | Flow30 | Flow60 |
| Average price               | 8.59     | 9.43   | 9.71   | 8.97     | 9.59   | 9.88   | 8.21      | 9.26   | 9.53   |
| $ P_t - P_{CE} $            | -1.21    | -0.37  | -0.09  | -0.83    | -0.21  | 0.08   | -1.59     | -0.54  | -0.27  |
| $ P_t - P_{t-1} $           | 0.14     | 0.15   | 0.17   | 0.14     | 0.15   | 0.19   | 0.14      | 0.15   | 0.16   |
| $\text{Std}(P_t - P_{t-1})$ | 0.08     | 0.05   | 0.06   | 0.08     | 0.05   | 0.07   | 0.07      | 0.05   | 0.06   |

*Note:* Summary statistics using prices from each second across 5 groups and 120 seconds over 20 trading periods.

## S2 Experiment Instructions

### S2.1 CDA

Welcome, and thank you for participating!

From now until the end of the experiment, please do not communicate with other participants. Please turn off your cell phones and stay focused on the task. If you have any questions, please raise your hand or let the experimenters know; they will answer your questions.

Please pay careful attention to the instructions, as real money is at stake. During the experiment, you will earn Experimental Currency Units (ECUs). At the end of the experiment, we will convert your earnings into US Dollars (USD) at one dollar for every 1000 ECUs. You are guaranteed a show-up fee of USD 6 but can earn considerably more.

#### Basic Idea

In this experiment, you and seven other participants will be traders in a simple automated financial market. Using the information displayed on your screen, you will make trades to earn as many ECUs as possible. As explained below, your earnings will depend on the trade offers you and the other traders make in your group. There will be 22 trading periods, each lasting 2 minutes.

The two first rounds are practice rounds (they don't count for your actual payment).

Your final earnings for the session will be the sum of your show-up fee and the cumulative profits from all trading periods, converted into US Dollars.

#### Buy Orders and Sell Orders (Treat CDA)

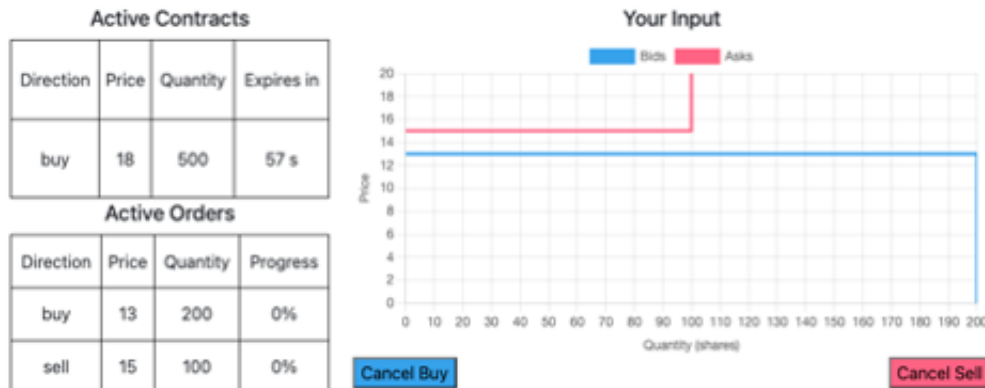
All traders can submit or update *buy* and/or *sell orders*. Each order consists of two elements or pieces of information:

- The ***total quantity*** stating the number of shares that the agent wants to trade.
- The ***limit price*** of the order. In buy orders, the limit price indicates the maximum price the trader is willing to pay per share. In sell orders, the limit price indicates the minimum price the trader is willing to accept for each share.

**Example** The blue line and blue square in Figure 10 show that the trader submitted a buy order for a total of 200 shares at a limit price of 13 ECUs. If someone is willing to sell for 13 ECUs or less, the former trader's buy order will be filled. If that counterpart wants to trade less than 200 shares, then the order will be filled only partially.

Figure 1 also shows (in red) that the trader submitted a sell order for 100 shares. The limit price of this order is 15 ECUs. That is, if someone is willing to buy at 15 ECUs or a higher price, the sell order will be filled. If said counterpart wants to trade less than 100 shares, then the order will be filled only partially.

Figure S10 CDA Your Input



**Submitting orders in the graphical interface (Treat CDA)** To set a sell order, you drag the red square anywhere inside the “your input” grid to set the limit price and the total quantity (volume). You submit your order by clicking on the “Send Sell” red button, as is shown in Figure 11.

Similarly, to set a buy order, you can drag the blue square anywhere inside the grid to set the limit price and the total quantity (volume). Then you can submit your order by clicking on the “Send Buy” button.

In case your order is not completely executed, your order will remain active with the remaining quantity. For example, if your order is for buying 200 shares and you only obtain 150 shares when trading with another participant. Then, your order remains active, but now only for 50 shares.

You can submit a new buy (sell) order, or cancel it, at any time. However, if you have an active order, you first need to cancel it by pressing the “cancel buy” (“cancel sell”) button. Then you can submit a new order.

Figure S11 CDA Your Input



## Contracts

Why do you want to trade? What is a share worth to you? The answer to both questions is that the computer will give you a contract that enables you to profit from trading. You will receive your

contract near the beginning of the trading period, and it will appear in the “Active Contracts” table on the left of the screen, as shown in Figure 10.

For example, your contract might read **“Buy 500 shares at a price of 18 ECUs in 97 seconds.”** This means that 97 seconds from now, the computer will buy up to 500 shares from your inventory at a price of 18. To take advantage of this opportunity, you need to buy some shares to build up your inventory. Your profit on each share (up to the 500th share) is the contract price (here 18) minus the price you paid when you bought that share. Of course, for this contract it would not be worthwhile to buy shares at a higher price than 18 or to buy more than 500 shares.

Similarly, a contract that says **“Sell 150 shares at a price of 13 ECUs in 80 seconds”** means that, in 80 seconds, the computer will sell you up to 150 shares at the price of 13 ECUs per share. Here you make a profit by selling up to 150 shares at prices above 13 ECUs per share. For this contract, it would not be worthwhile to sell shares at a lower price than 13 or to sell more than 150 shares.

Keep in mind that you can buy more shares than you have cash for and thus have a negative cash balance, or sell more shares than you own and thus have a negative inventory (this is called shorting). If you have a BUY contract for 400 shares, your inventory should be between zero and your contract quantity, 400 shares. If you have a SELL contract for 500 shares, your inventory should be between -500 (minus 500) shares and zero.

### **How the market works (Treat CDA)**

The market price at which you can buy or sell shares depends on other traders' buy and sell orders. Consider the example illustrated in Figure 12. All traders, except Trader 1, have active buy or sell orders. In particular, the four sellers want to sell at prices of 15 or more, respectively (see red steps). These four sellers conform to the market supply (the sum of their four sell orders) shown in red. The four buyers want to buy at prices of 13, 12 and 11 or less, respectively (see blue steps). These three buyers conform to the market demand (the sum of their three buy orders) shown in blue.

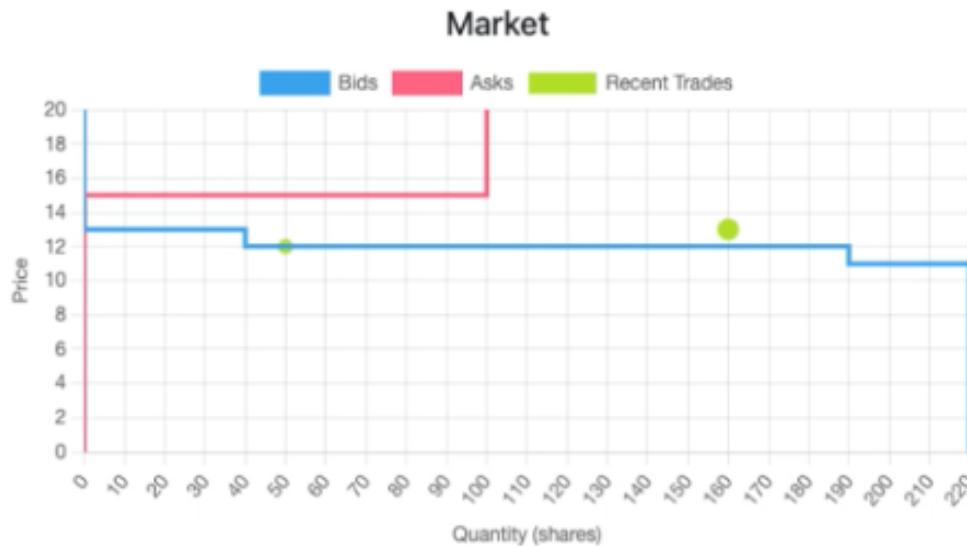
The green dots indicate the most recent trades. The latest trade is represented by the largest dot.

As you can see, the demand and supply do not cross each other inside the box but along the y-axis. That is there cannot be trade at that point.

Let us see what happens when Trader 1 decides to submit a buy order. There are two cases.

- First, if her buy order has a price below the lowest sell order price (say 15), then Trader 1 will only add to the demand, and no transaction will occur.
- Second, if Trader 1's buy order comes at a price above the lowest selling price (say 13), then she will transact immediately with the seller(s) with the lowest price. Suppose the seller with the lowest price does not offer enough volume (quantity) to fill Trader's 1 order. In that case, the buy order of Trader 1 will be filled only partially or the remaining quantity will be purchased from the second lowest selling price, and so on.

Figure S12 CDA Market



## Your Earnings

You make a profit by selling shares at higher prices than you bought them.

However, notice that you do not need to buy a share in order to sell it. Even if you have zero inventory you can still sell shares (making your inventory negative). Similarly, even if you have no cash, you can still buy shares (making your cash balance negative).

Also, notice that you can buy and sell from and to either the computer and other participant traders, or both. Your profits from transactions with other participants will appear immediately in your cash account, whereas your profits from contracts with the computer will be added to the cash account when the contract expires (aka contract deadline).

**Examples:** You lose money when you buy at a higher price than you sold, or sell at a lower price than you bought. For example, suppose you pay 23 ECUs for one share that you sell to the computer at a contract price of 17 ECUs. Then your profit is  $17 - 23 = -6$  ECUs. That is, you incurred a loss of 6 ECUs on that share. If, instead, you bought that share at price of 14 ECUs before the deadline, you would get a profit of  $17 - 14 = 3$  ECUs on that share.

### Projected Profits

The middle bottom box shows your *projected profits* which indicate your period profits if the active contract (if any) expires immediately and the period ends right away. This provides a summary of your current state.

**Session Dollar Earnings:** After the last trading round, the computer will add up all your period earnings to determine your actual compensation. These earnings will be converted into US dollars at one dollar for every 1000 ECUs.

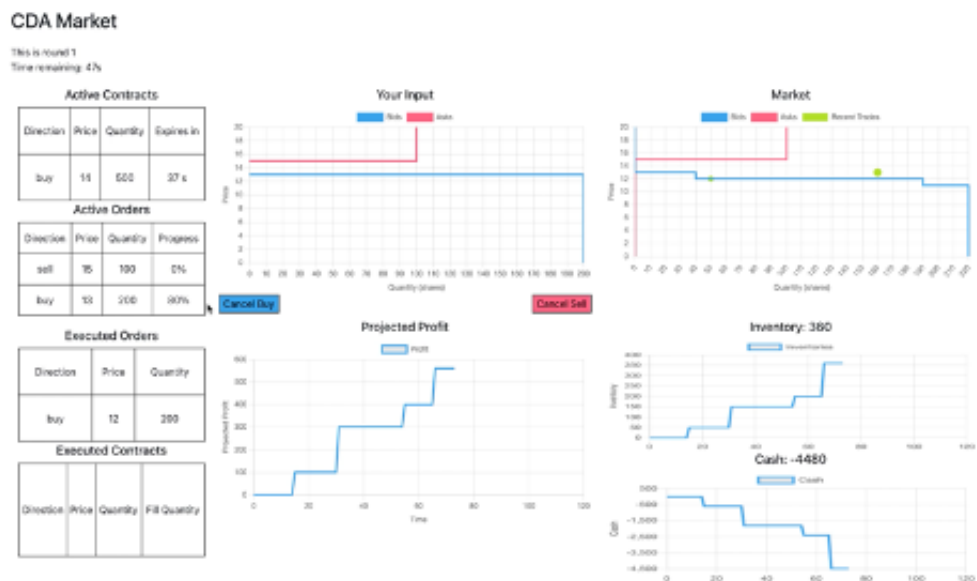


Figure S13 CDA UI

## S2.2 FLOW

Welcome, and thank you for participating!

From now until the end of the experiment, please do not communicate with other participants. Please turn off your cell phones and stay focused on the task. If you have any questions, please raise your hand or let the experimenters know; they will answer them.

Please pay careful attention to the instructions, as real money is at stake. During the experiment, you will earn Experimental Currency Units (ECUs). At the end of the experiment, we will convert your earnings into US Dollars (USD) at one dollar for every 1000 ECUs. You are guaranteed a show-up fee of USD 6 but can earn considerably more.

### Basic Idea

In this experiment, you and seven other participants will be traders in a simple automated financial market. Using the information displayed on your screen, you will make trades to earn as many ECUs as possible. As explained below, your earnings will depend on the trade offers you and the other traders make in your group. There will be 20 trading periods, each lasting 2 minutes.

The first two rounds are practice rounds (they don't count for your actual payment).

Your final earnings for the session will be the sum of your show-up fee and the cumulative profits from all trading periods, converted into US Dollars.

### Buy Orders and Sell Orders (Treat FLOW)

All traders can submit or update *buy* and/or *sell orders*. Each order consists of four elements:

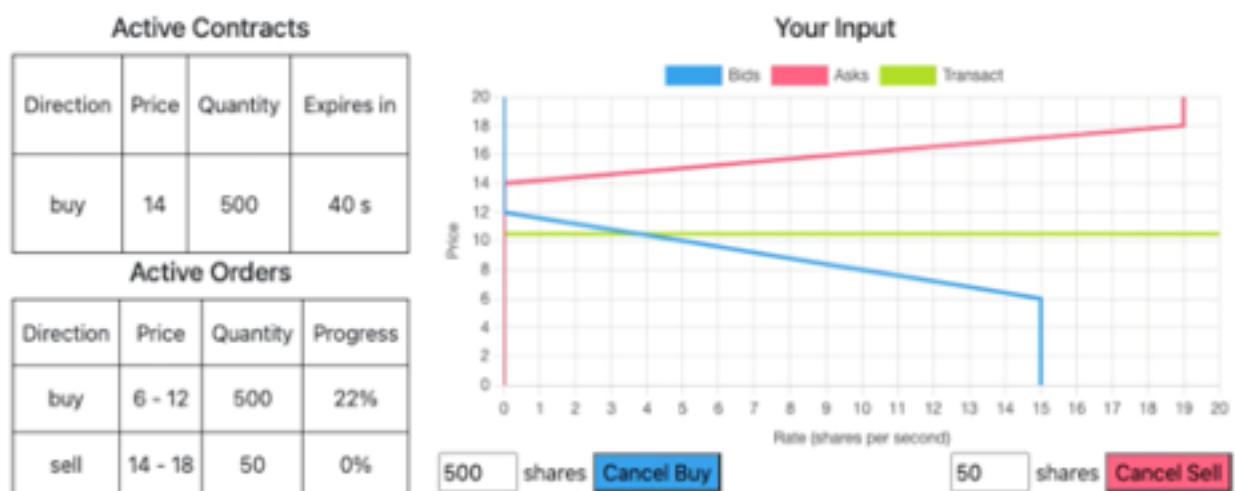
- The **total quantity** stating the number of shares that the agent wants to trade,

- The **maximum rate** of shares to trade per second.
- The **low price** of the order. In buy orders, the low price indicates the price under which the order is filled at the *maximum rate*. In sell orders, the low price indicates the price under which the order is not filled.
- The **high price** of the order. In buy orders, the high price indicates the price above which the order is not filled. In sell orders, the high price indicates the price above which the order is filled at the *maximum rate*.

**Example** The blue line and blue box in Figure 14 show that the trader submitted a buy order for a total of 500 shares at the maximum rate of 15 shares per second at prices between 6 (low price) and 12 (high price). When the market price is at or below 6 ECUs per share, her order will be filled at her *maximum rate* (15 shares per second). Her buy order will not be executed when the market price is at or above 12 ECUs per share. When the market price is between 6 to 12 ECUs per share, her order will be filled at a rate between zero and her maximum rate. The higher the market price, the lower the rate of execution.

Figure 14 also shows (in red) that the trader submitted a sell order for 50 shares. Her *maximum rate* is 19 shares per second, her *low price* is 14, and her *high price* is 18. When the market price is at or above 18 ECUs per share, her order will be filled at her *maximum rate* of 19 shares per second. Her sell order will not be executed when the market price is at or below 14 ECUs per share. When the market price is between 14 to 18 ECUs per share, her order will be filled at a rate between zero and her maximum rate. The higher the market price, the higher the rate of execution.

Figure S14 FLOW Your Input



**Submitting orders in the graphical interface (Treat FLOW)** To set a sell order, you move the red dot along the vertical axis to set the low price, and you drag the red square anywhere inside the grid to set the high price and maximum rate. You enter the **total number of shares** you want to



sell in the box next to the “Send Sell” red button. When you have finished setting the order, submit your order by clicking on the “Send Sell” button.

Similarly, to set a buy order, you move the blue dot along the vertical axis to set the high price, and you drag the blue square anywhere inside the grid to set the low price and the maximum rate. You enter the **total number of shares** you want to buy in the box next to the “Send Buy” blue button. When you have finished setting the order, submit your order by clicking on the “Send Buy” button.

You can submit a new buy (sell) order, by first canceling your active buy (sell) orders and then submitting a new buy (sell) order. You do not need to wait for an order to be fully executed to cancel it.

## Contracts

Why do you want to trade? What is a share worth to you? The answer to both questions is that the computer will give you a contract that enables you to profit from trading. You will receive your contract near the beginning of the trading period, and it will appear in the “Active Contracts” table on the left of the screen, as shown in Figure 14.

For example, your contract might read **“Buy 500 shares at a price of 14 ECUs in 40 seconds.”** This means that 40 seconds from now, the computer will buy up to 500 shares from your inventory at a price of 14. To take advantage of this opportunity, you need to buy some shares to build up your inventory. Your profit on each share (up to the 500th share) is the contract price (here 14) minus the price you paid when you bought that share. Of course, for this contract it would not be worthwhile to buy shares at a higher price than 14 or to buy more than 500 shares.

Similarly, a contract that says **“Sell 150 shares at a price of 13 ECUs in 80 seconds”** means that, in 80 seconds, the computer will sell you up to 150 shares at the price of 13 ECUs per share. Here you make a profit by selling up to 150 shares at prices above 13 ECUs per share. For this contract, it would not be worthwhile to sell shares at a lower price than 13 or to sell more than 150 shares.

Keep in mind that you can buy more shares than you have cash for and thus have a negative cash balance, or sell more shares than you own and thus have a negative inventory (this is called shorting). If you have a BUY contract for 400 shares, your inventory should be between zero and your contract quantity, 400 shares. If you have a SELL contract for 500 shares, your inventory should be between -500 (minus 500) shares and zero.

## How the market works (Treat FLOW)

The market price and the rate at which you can buy or sell shares depends on the buy and sell orders of all traders, as illustrated in the top chart of Figure 15. Here, one trader submitted the buy order and sell order shown in Figure 14. Together with the buy and sell orders submitted by the other 7 traders, they resulted in the market demand (sum of buy orders) shown in blue and market supply (summed sell orders) shown in red. The supply and demand intersect at a price of 10.48 and a Rate of 3.8. Therefore the trader from Figure 14 is buying at a price of 10.48 ECU to some other trader(s) at a rate of 3.8 shares per second (where the price crosses the agents’ demand curve). Thus, her cash position is decreasing at a rate of  $10.48 \times 3.8 = 19.82$  ECU per second while her inventory is increasing at a rate of 3.8 shares per second. This will continue until she buys her total